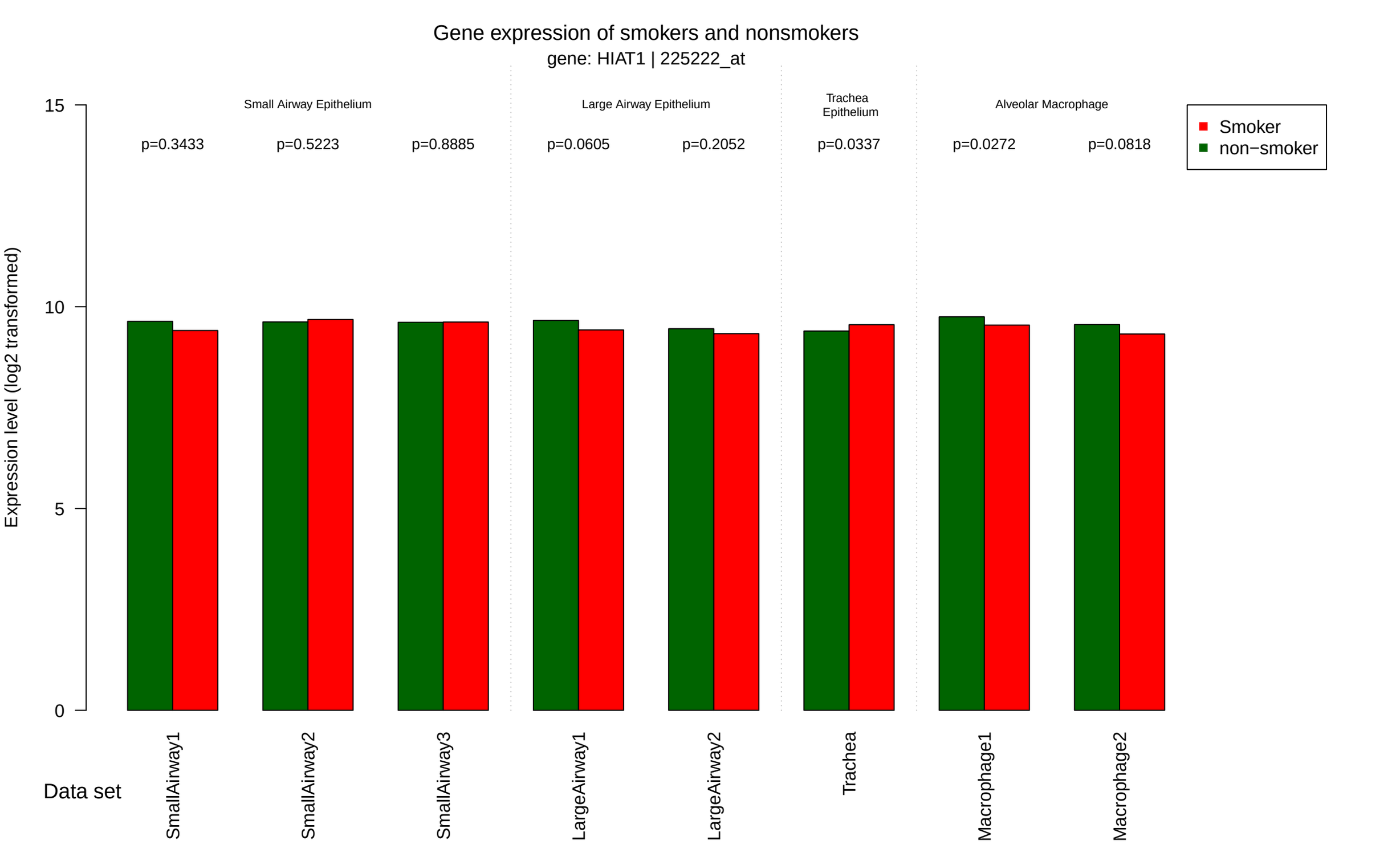
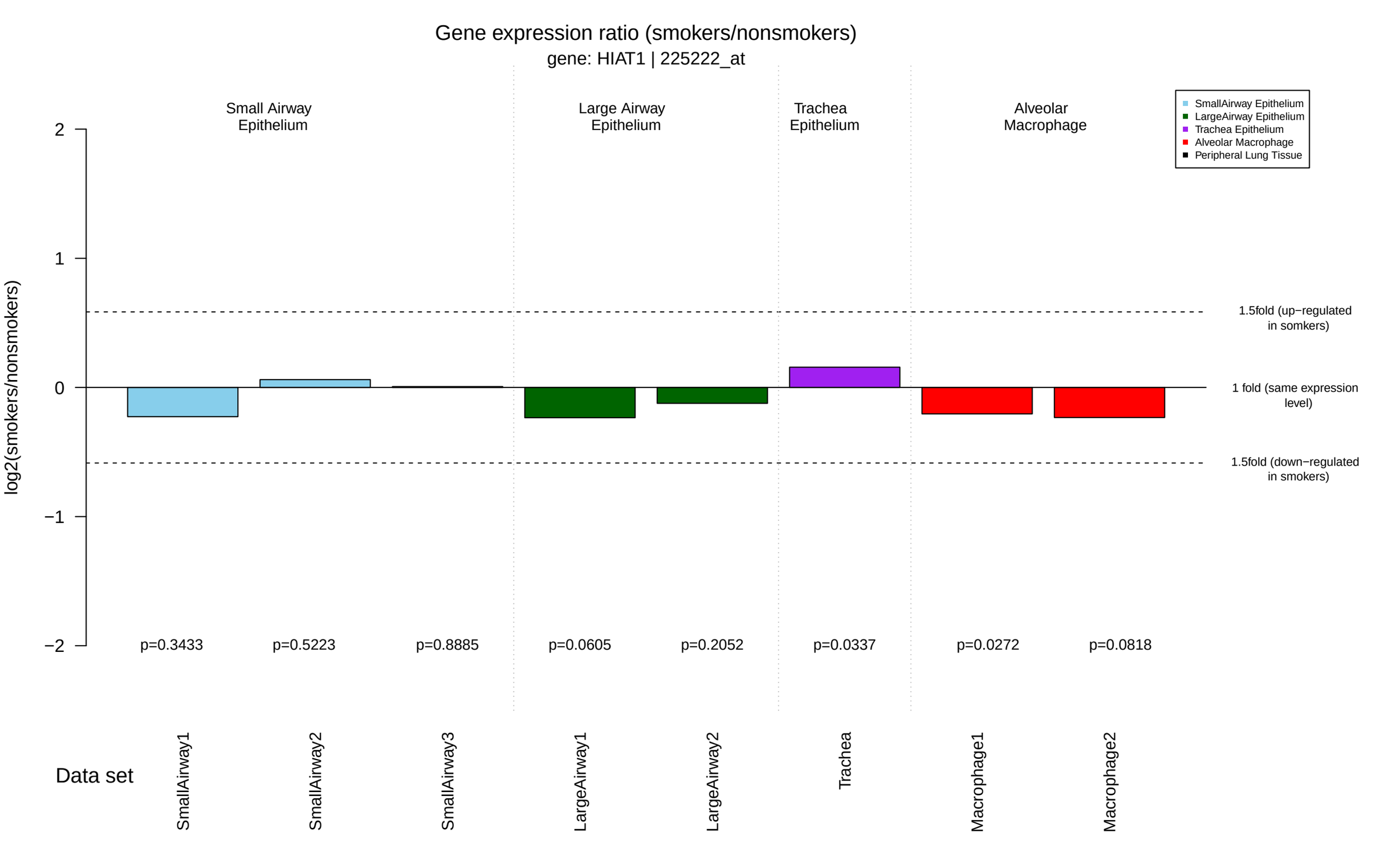
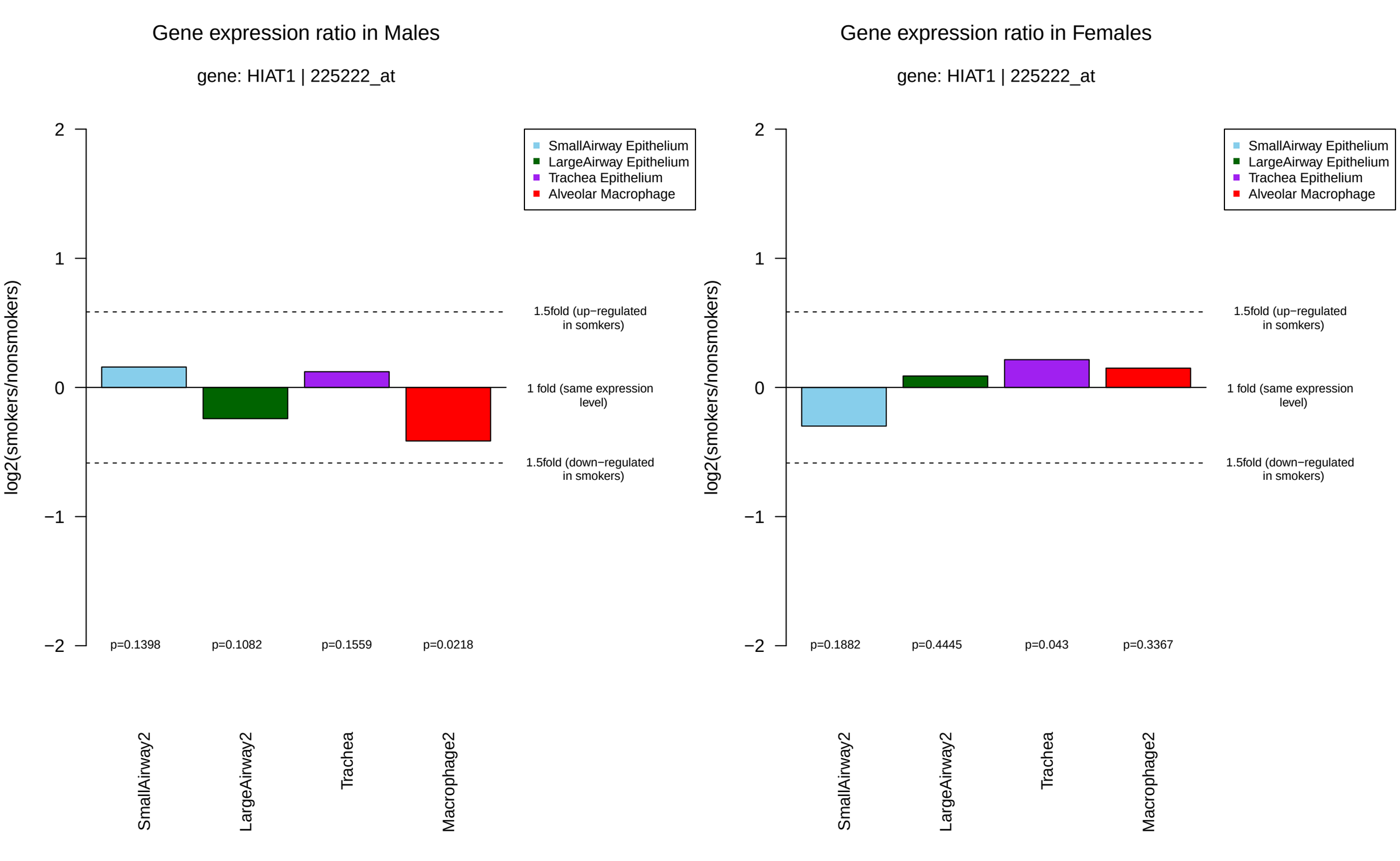
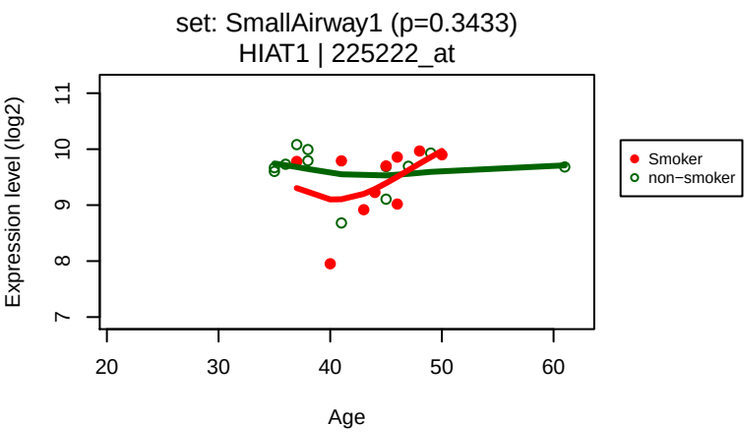
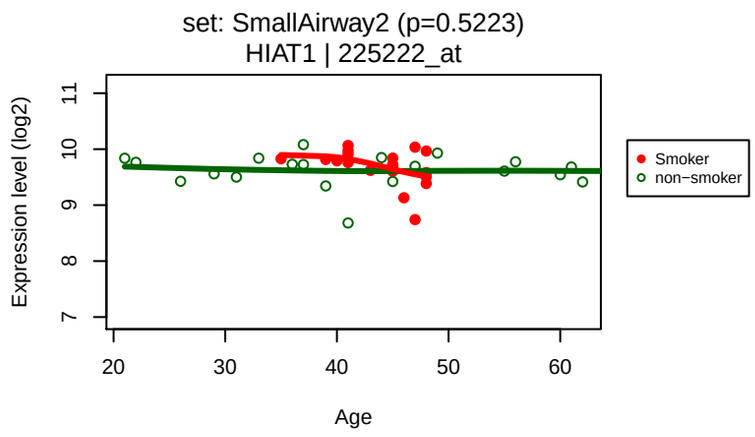
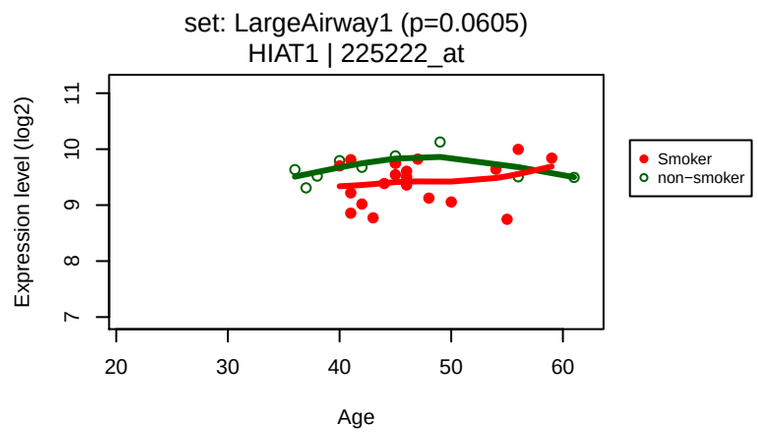

Gene expression of smokers and nonsmokers
gene: HIAT1 | 225222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3433
p=0.5223
p=0.8885
p=0.0605
p=0.2052
p=0.0337
p=0.0272
p=0.0818
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

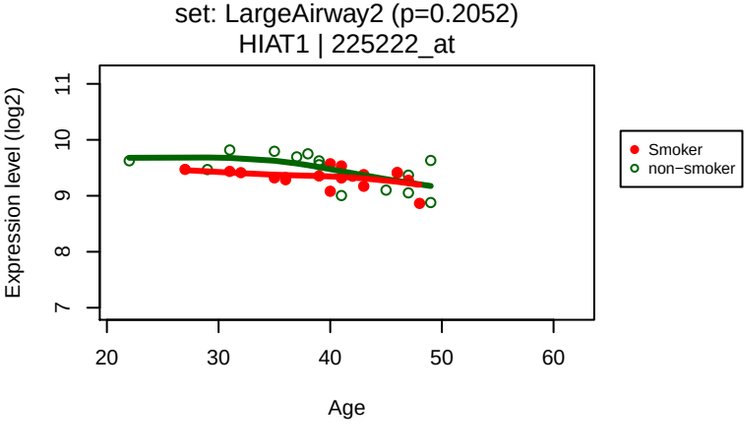

Gene expression ratio (smokers/nonsmokers)
gene: HIAT1 | 225222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3433
p=0.5223
p=0.8885
p=0.0605
p=0.2052
p=0.0337
p=0.0272
p=0.0818
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

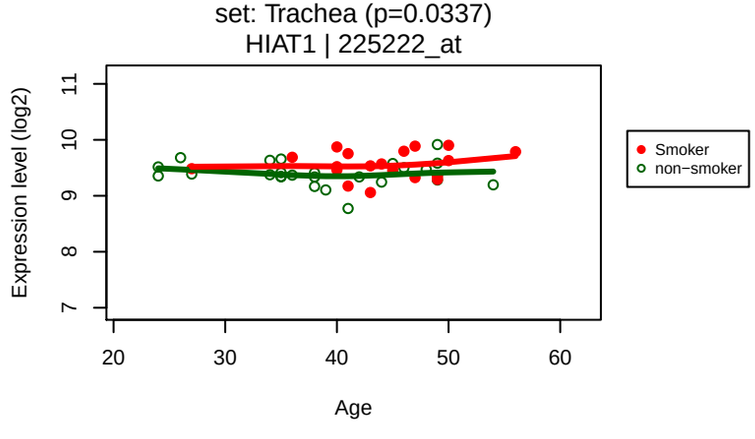

Gene expression ratio in Males
gene: HIAT1 | 225222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1398
p=0.1082
p=0.1559
p=0.0218
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIAT1 | 225222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1882
p=0.4445
p=0.043
p=0.3367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

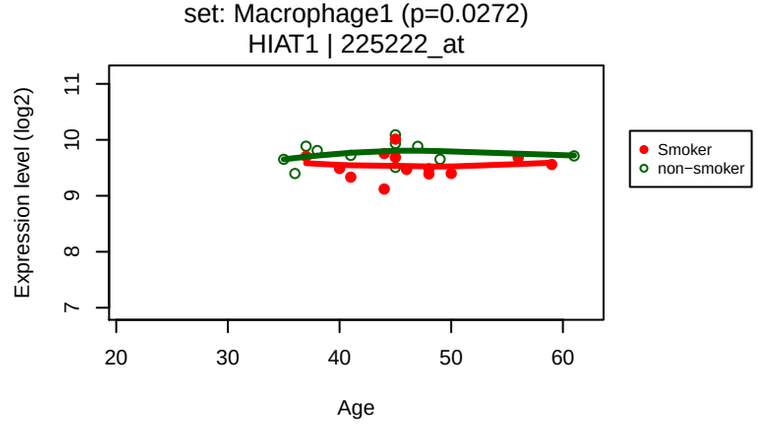

set: SmallAirway1 (p=0.3433)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker

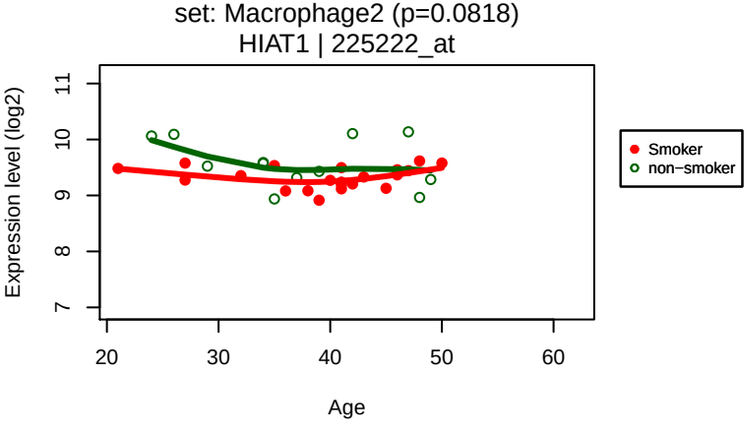

set: SmallAirway2 (p=0.5223)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker

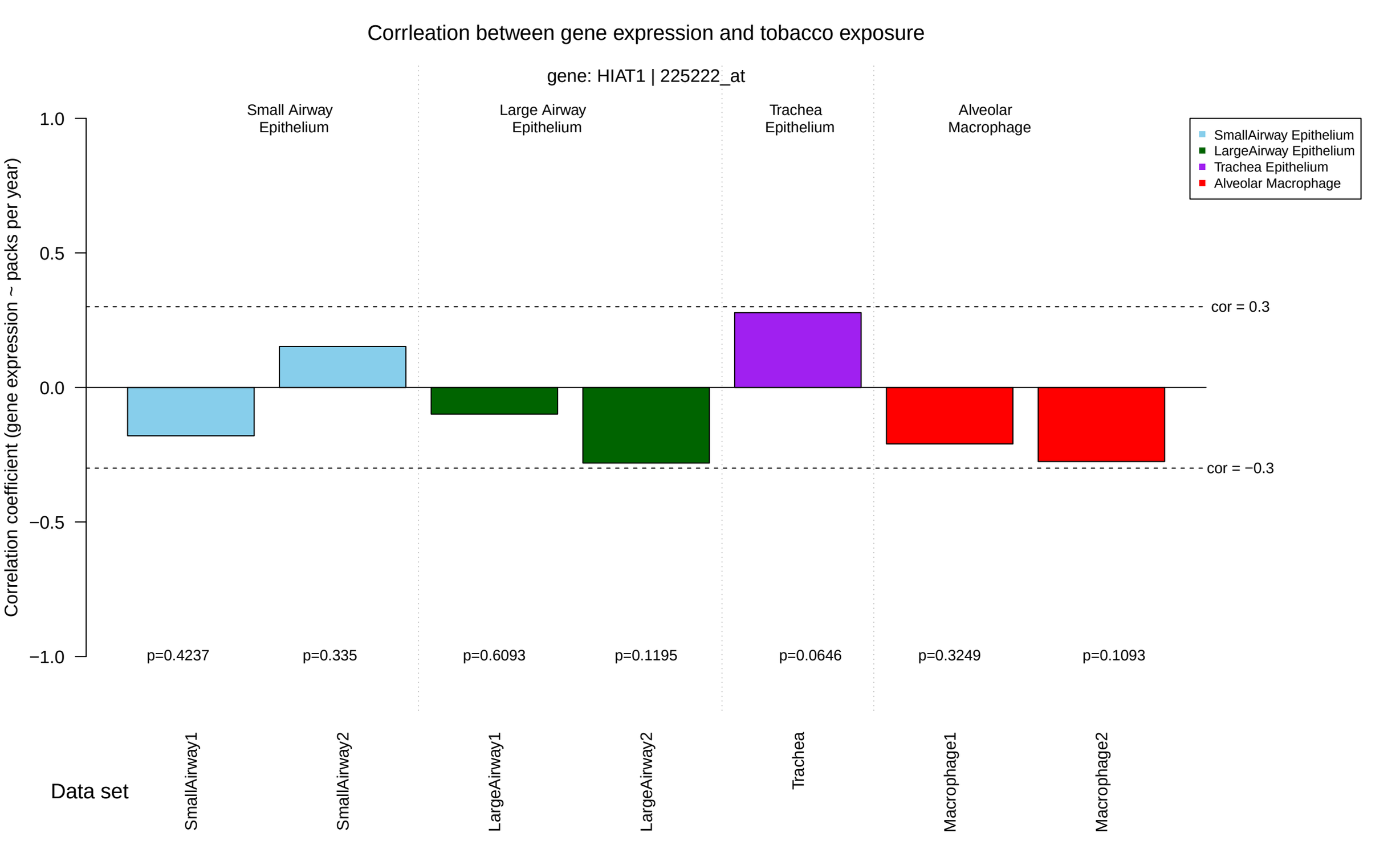

set: LargeAirway1 (p=0.0605)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2052)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0337)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0272)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0818)
HIAT1 | 225222_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIAT1 | 225222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4237
p=0.335
p=0.6093
p=0.1195
p=0.0646
p=0.3249
p=0.1093
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2